TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders
Standard → slope-intercept: subtract the x-term, then divide every term by the number with y.
Point-slope → slope-intercept: distribute, then add or subtract to get y alone.
Intercepts: x-intercept: put 0 in for y. y-intercept: put 0 in for x.
Solve a formula for a letter: undo everything around that letter, doing the same to both sides.

1. What value of x makes $2(x + 3) = x - 4$ true?

Tip: distribute, then get [[x]] on one side

A. -10

B. -7

C. 2

2. How many solutions does $3x + 1 = 3x - 5$ have?

Tip: same [[x]]-term + different numbers → no solution

F. One solution, $x = 0$

G. No solution

H. Infinite solutions

3. What are the slope and y-intercept of $y = -2x + 7$?

> **Tip:** [[y = mx + b]]: slope [[m]], [[y]]-intercept [[(0, b)]]

A. slope 7, $(0, -2)$

B. slope 2, $(0, 7)$

C. slope -2, $(0, 7)$

4. What is the x-intercept of $y = 2x - 6$?

> **Tip:** [[x]]-intercept: put 0 in for [[y]]

F. $(-6, 0)$

G. $(0, -6)$

H. $(3, 0)$